BLUESTOCK
MUTUAL FUND ANALYTICS

CAPSTONE PROJECT

An End-to-End Data Analytics Pipeline for the Mutual Fund Industry

Submitted By

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Technologies Used

Python • Pandas • SQLite • SQL • Power BI • GitHub

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Abstract

This project presents a comprehensive analysis of the mutual fund industry through a complete data analytics pipeline. The workflow spans data ingestion, data cleaning, exploratory data analysis, performance analytics, advanced analytics, and dashboard development.

The primary objective is to transform raw mutual fund data into actionable insights that support a deeper understanding of investment trends, fund performance, and overall market behaviour, ultimately enabling more informed, data-driven decision making for investors and stakeholders.

Project Objectives

The key objectives defined for this capstone project are as follows:

- Build an end-to-end ETL (Extract, Transform, Load) pipeline for mutual fund data.
- Clean and validate the underlying datasets to ensure data quality and consistency.
- Perform Exploratory Data Analysis (EDA) to uncover patterns and trends.
- Calculate key fund performance metrics for risk and return assessment.
- Conduct advanced analytics to derive deeper, actionable insights.
- Build an interactive Power BI dashboard for stakeholder reporting.
- Generate clear business insights to support investor decision-making.

Dataset Description

The analysis is built on ten interconnected datasets covering different aspects of the mutual fund industry, ranging from fund-level reference data to investor transaction records.

No.	Dataset	Description
1	Fund Master	Core reference details for each mutual fund scheme.
2	NAV History	Historical Net Asset Value records used to track fund movements over time.
3	AUM by Fund House	Assets Under Management data segmented by fund house.
4	Monthly SIP Inflows	Monthly Systematic Investment Plan contribution data.
5	Category Inflows	Investment inflow data segmented by fund category.
6	Industry Folio Count	Investor folio counts across the mutual fund industry.
7	Scheme Performance	Performance records for individual mutual fund schemes.
8	Investor Transactions	Transaction-level records of investor activity.
9	Portfolio Holdings	Holdings data describing each fund's underlying portfolio.
10	Benchmark Indices	Benchmark index data used for performance comparison.

Total Datasets: 10 CSV files
Database: SQLite Database – bluestock_mf.db

Project Architecture

ETL Pipeline

The data pipeline follows a standard Extract, Transform, Load (ETL) approach, supported by a dedicated validation stage to ensure data reliability.

Extract

Raw CSV files are loaded using Pandas.

Transform

- Missing value handling
- Data type conversion
- Duplicate removal
- Validation checks

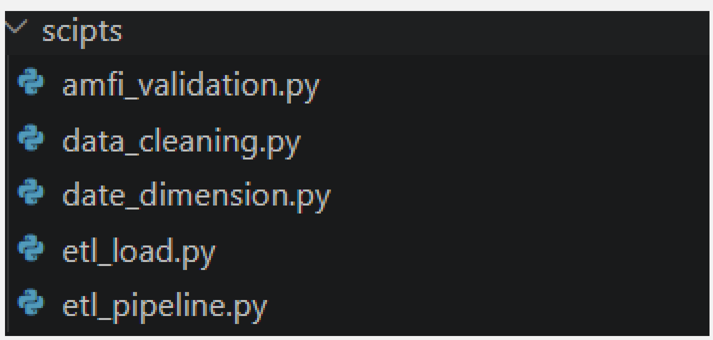
Load

Processed datasets are loaded into the SQLite database.

Validation

- AMFI code verification
- Missing value checks
- Schema validation

Supporting Screenshots



The project follows a modular ETL architecture. Separate scripts were developed for validation, cleaning, loading, and pipeline execution, improving maintainability and scalability.

Data Cleaning Process

Before analysis, all raw datasets were processed through a structured cleaning workflow to ensure accuracy, consistency, and analytical readiness.

Cleaning Activities

- Missing value treatment
- Duplicate record removal
- Date standardization
- Column normalization
- Data consistency checks

Output

Cleaned datasets are stored in the data/processed/ directory, ready for downstream analysis and reporting.

Exploratory Data Analysis

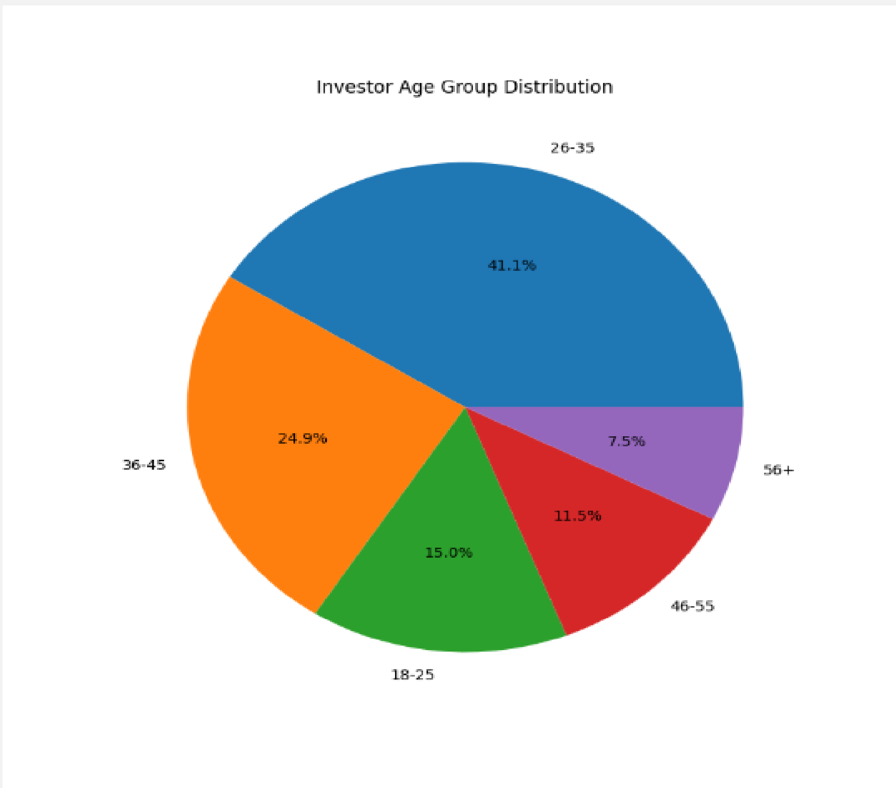
Exploratory Data Analysis (EDA) was carried out to understand the underlying structure of the data and to identify early trends across investors and fund performance.

Analyses Performed

- State-wise investor distribution
- Gender distribution
- AUM growth trend
- SIP trend analysis
- Category inflow analysis

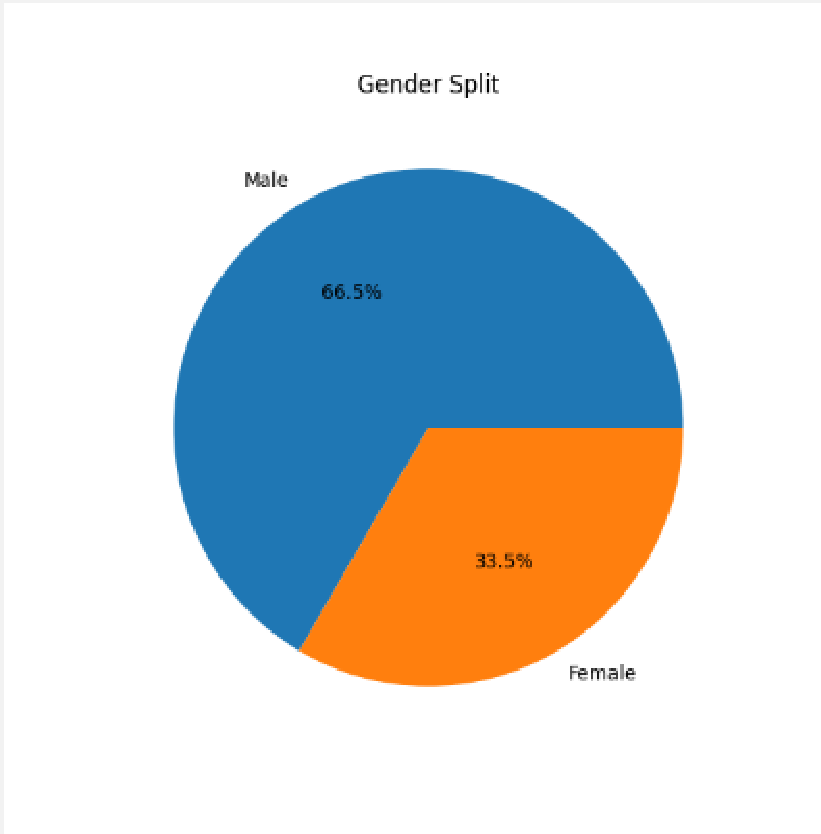
Supporting Visuals

INVESTOR AGE GROUP DISTRIBUTION



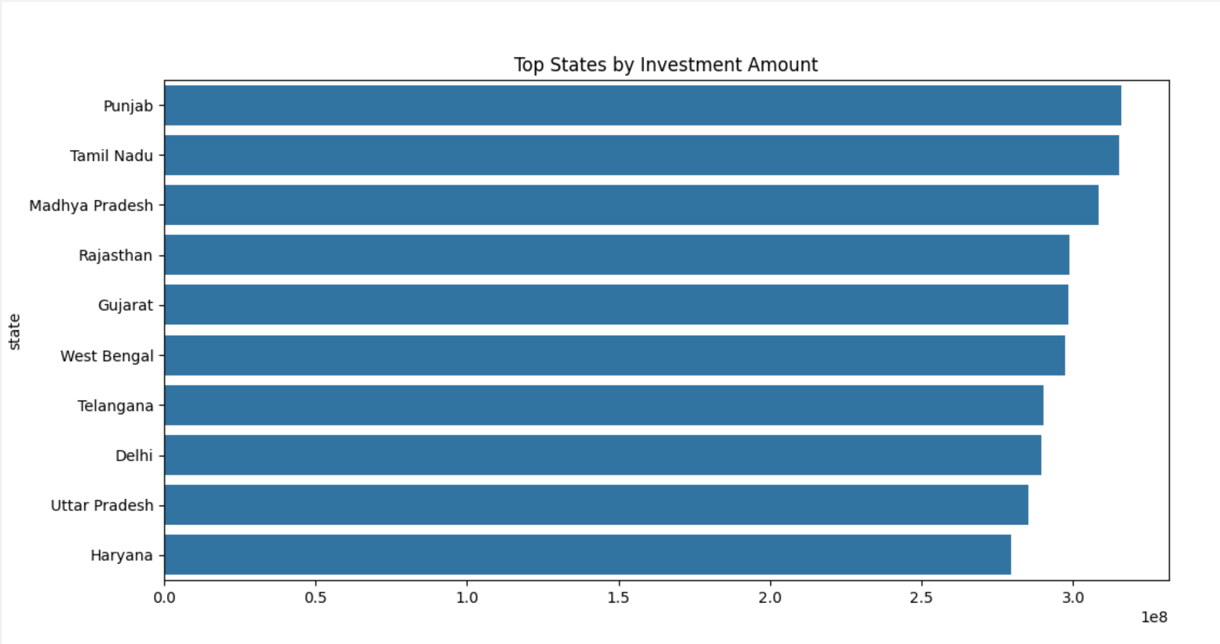
Findings: The majority of investors belong to the working-age population, indicating strong participation from individuals in their earning years. Younger investors are increasingly adopting mutual fund investments

Gender split



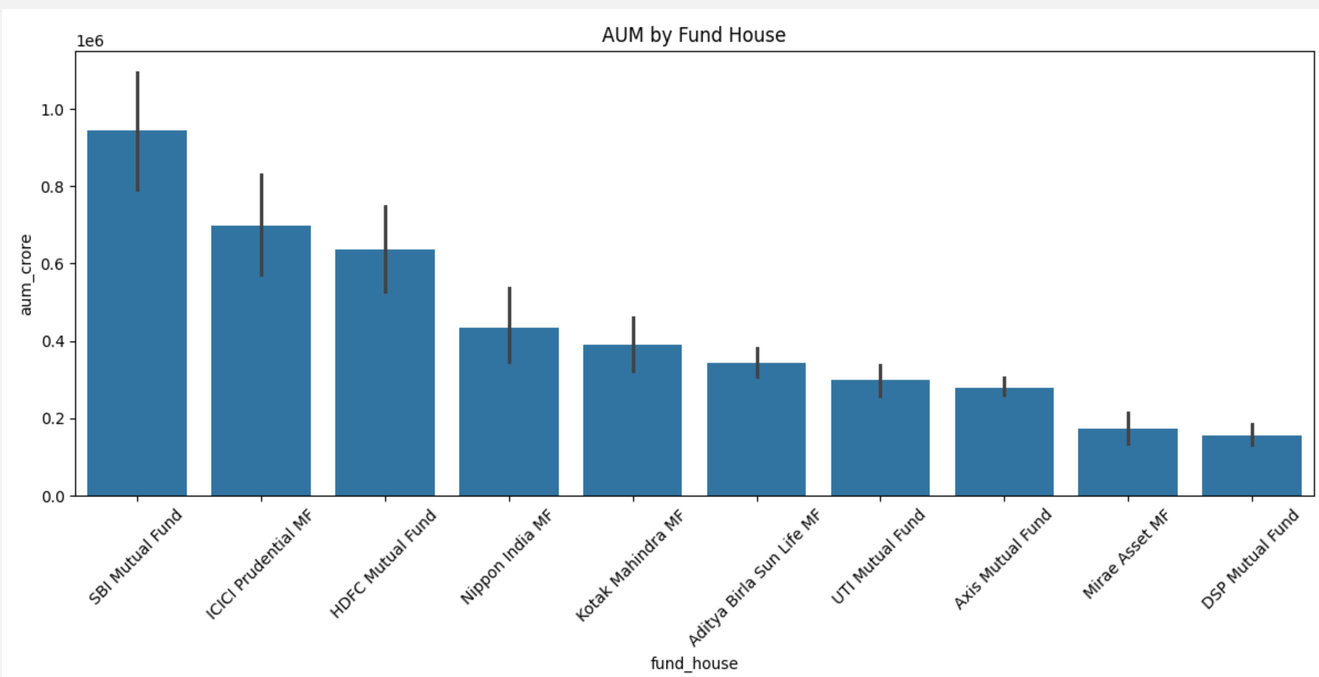
Findings: Male investors constitute a larger share of the investor base, though female participation is also significant, reflecting growing financial awareness.

TOP STATES BY INVESTMENT AMOUNT



Findings: Investor participation is concentrated in a few major states, highlighting regional differences in mutual fund adoption.

AUM BY FUND HOUSE



Findings: Assets Under Management have shown a consistent upward trend, indicating increasing investor confidence and growth in the mutual fund industry.

Performance Analytics

A set of standard performance metrics was calculated to evaluate fund returns on a risk-adjusted basis and to compare fund performance against relevant benchmarks.

Metrics Calculated

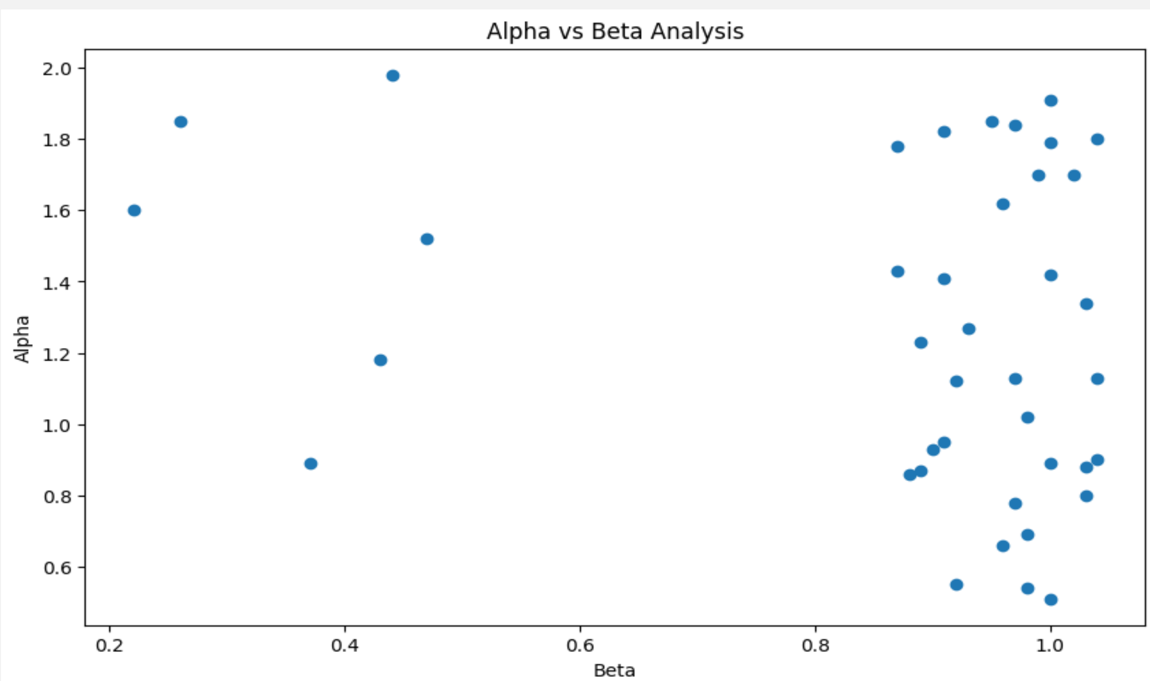
Metric	Description
CAGR	Compound Annual Growth Rate – the annualized rate of return over a period.
Alpha	Excess return of a fund relative to its benchmark.
Beta	Sensitivity of a fund's returns to overall market movements.
Volatility	Degree of variation in a fund's returns over time.
Sharpe Ratio	Risk-adjusted return measure comparing excess return to volatility.
Maximum Drawdown	Largest observed decline from a peak to a trough in fund value.

Purpose

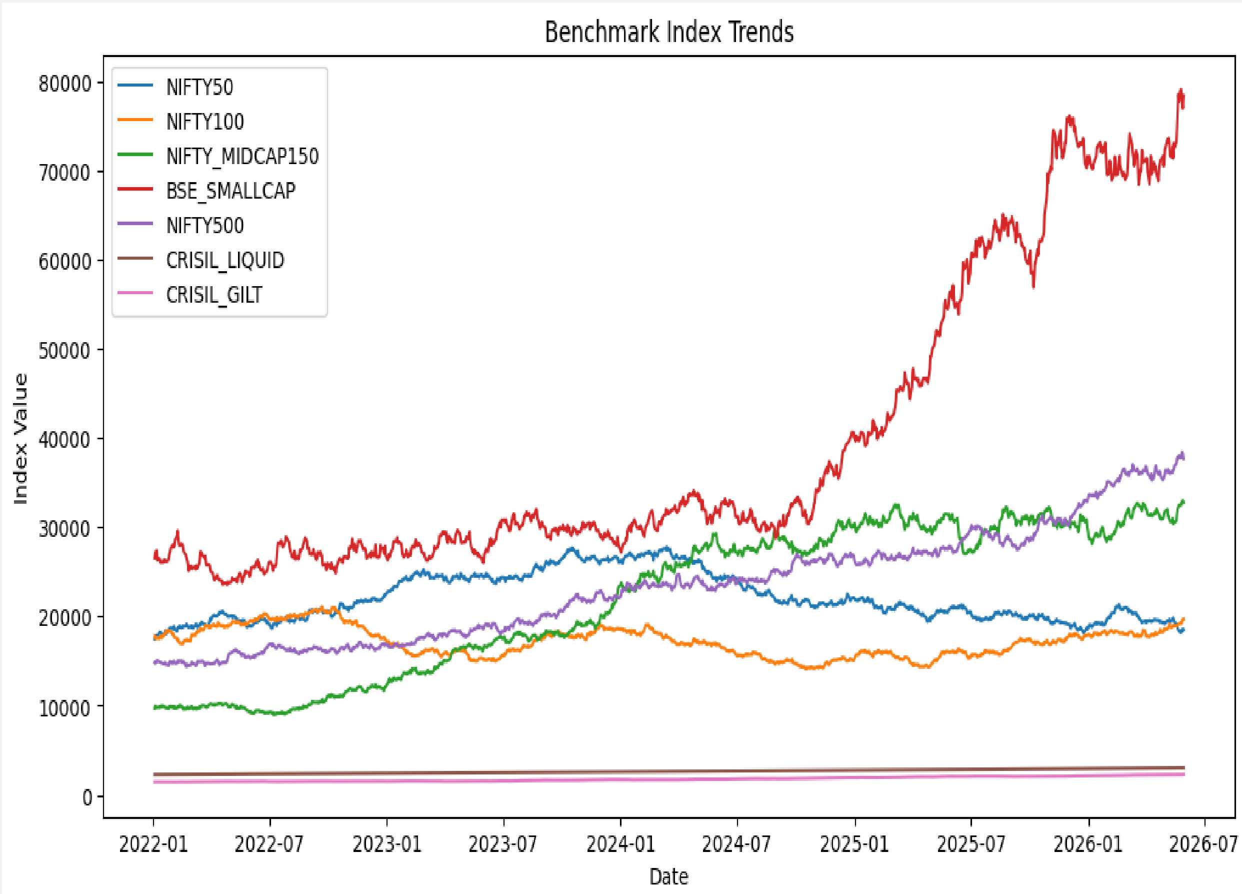
These metrics evaluate risk-adjusted fund performance and enable meaningful comparison against benchmark indices.

Supporting Outputs

ALPHA VS BETA ANALYSIS



BENCHMARK INDEX TRENDS



Performance metrics such as CAGR, Alpha, Beta, Sharpe Ratio, Volatility, and Maximum Drawdown were calculated to evaluate fund returns and risk-adjusted performance. Benchmark comparisons provide additional context for fund evaluation.

Advanced Analytics

Beyond standard performance metrics, additional advanced analyses were performed to uncover relationships between funds and identify top opportunities for investors.

Correlation Analysis

Examines the relationship among mutual fund NAVs to identify funds that move together or independently of one another.

Fund Ranking

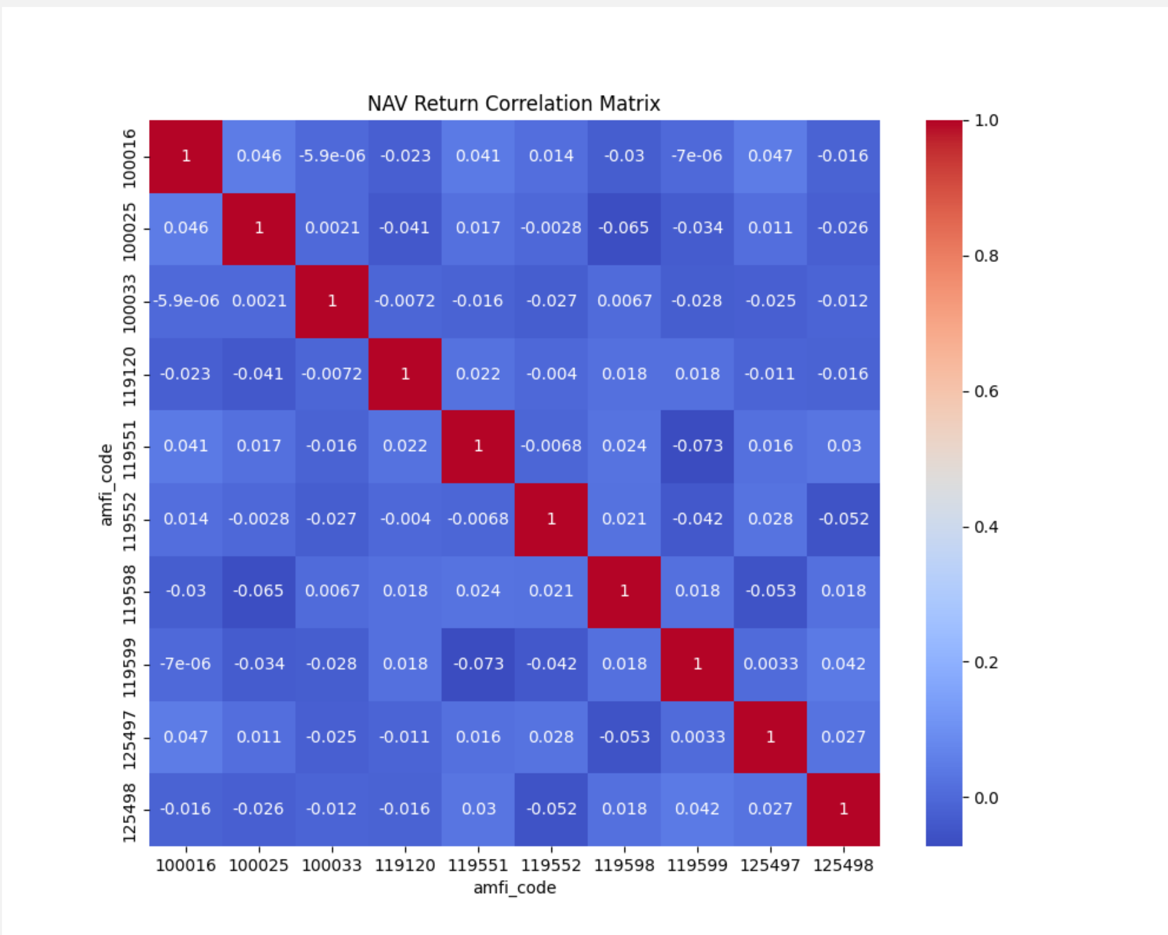
Identifies the top-performing mutual funds based on calculated performance metrics.

Sector Allocation Analysis

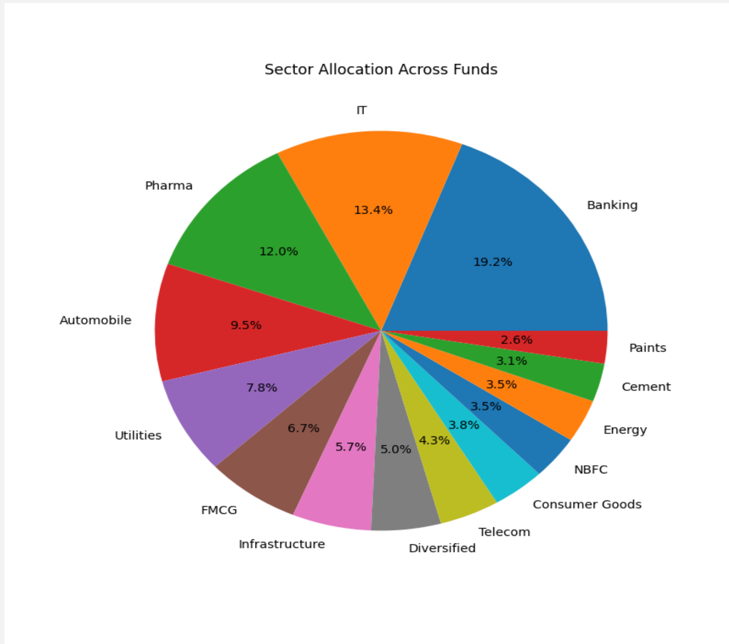
Analyzes sector-wise investment concentration to highlight where fund portfolios are most heavily weighted.

Supporting Visuals

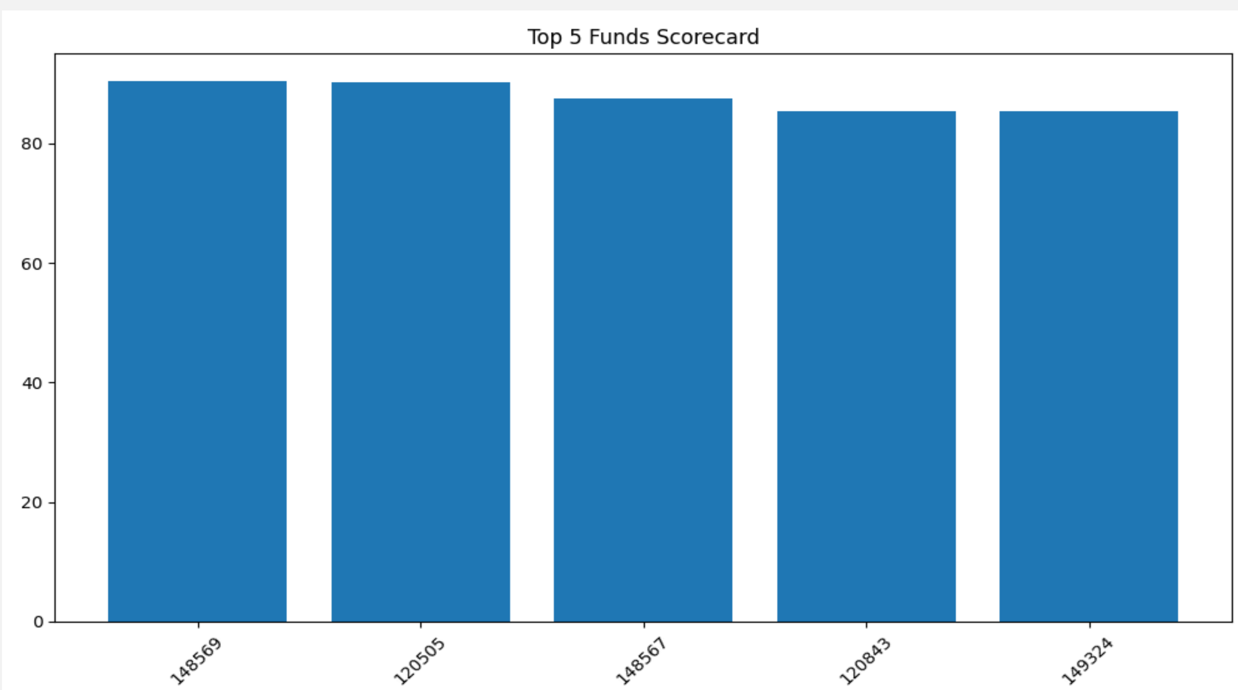
NAV RETURN CORRELATION MATRIX



SECTOR ALLOCATION ACROSS FUNDS



TOP 5 FUNDS SCORECARD



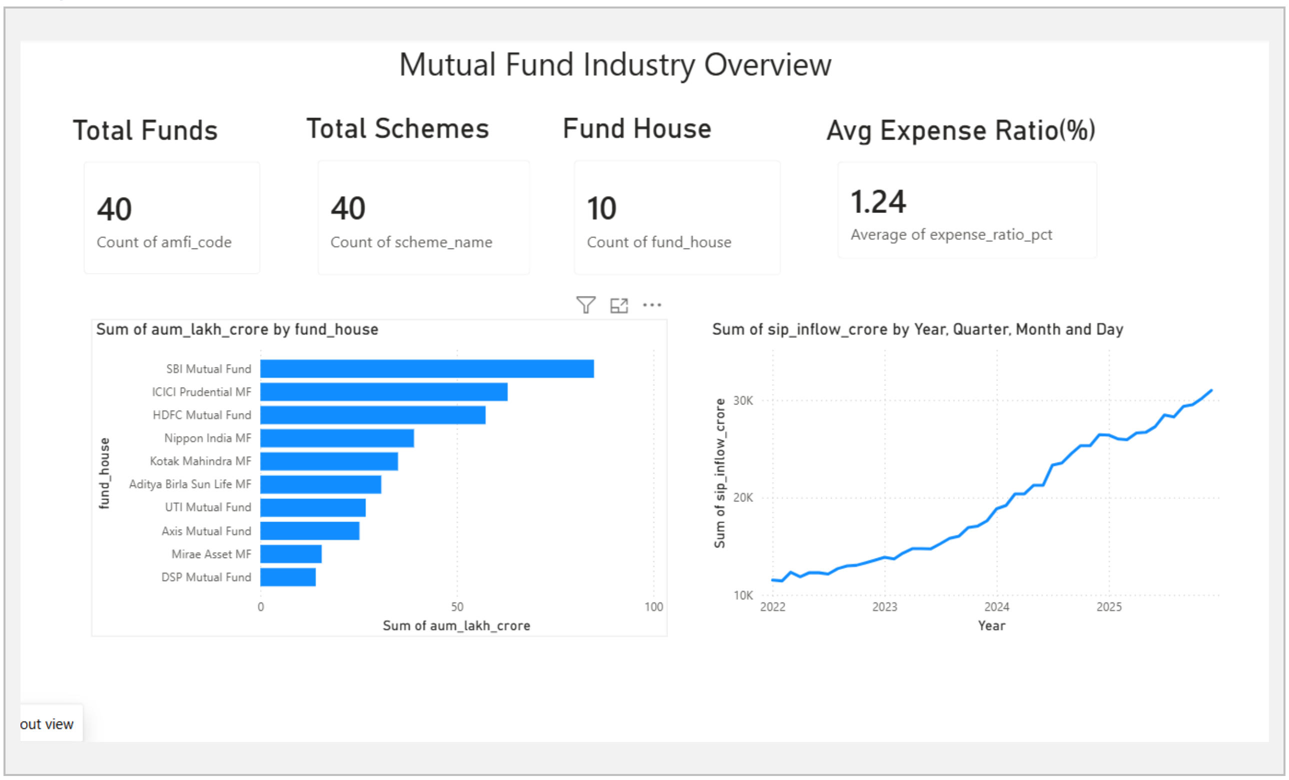
Power BI Dashboard

An interactive Power BI dashboard was built to present the analysis in a stakeholder-friendly format, organized across five dedicated pages.

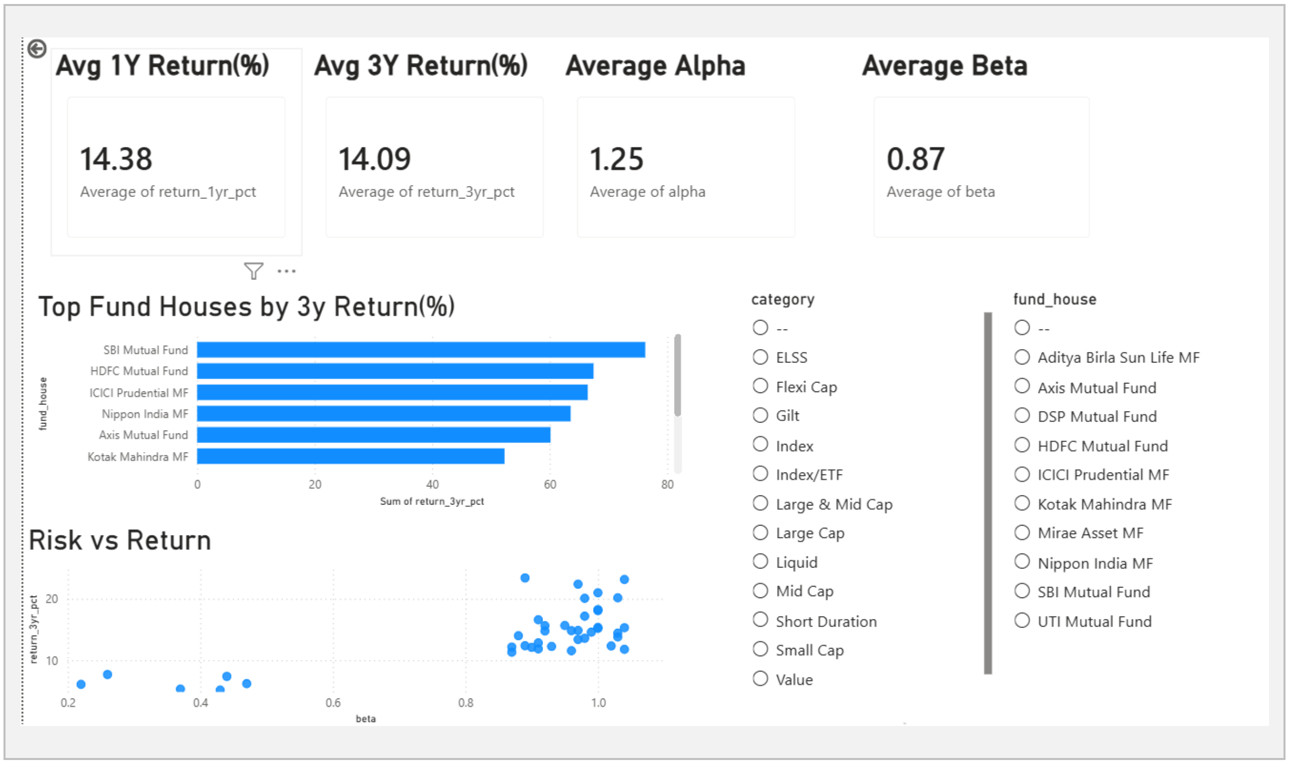
Page	Dashboard Section
1	Industry Overview
2	Fund Performance
3	Investor & SIP Analysis
4	Market Trends
5	Executive Summary

Supporting Screenshots

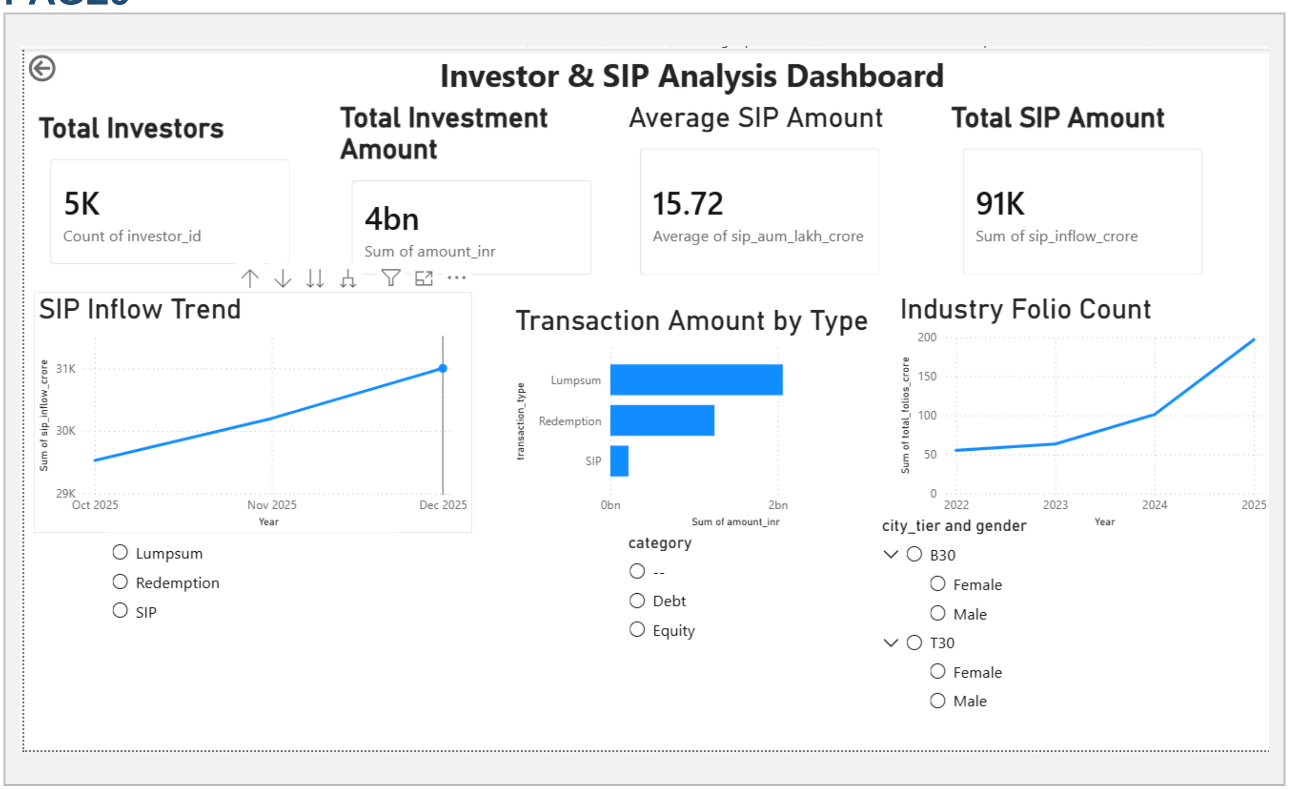
PAGE1



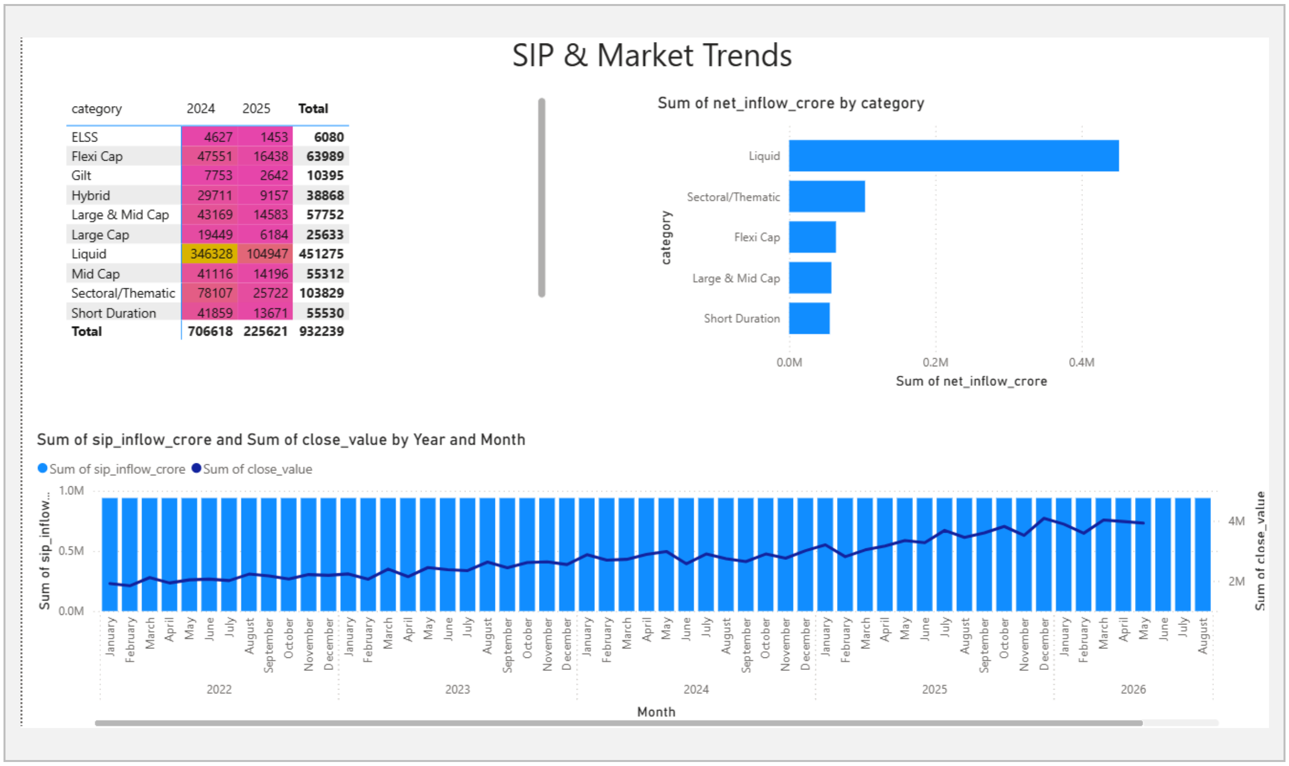
PAGE 2


Avg 1Y Return(%)
14.38
Average of return_1yr_pct
Avg 3Y Return(%)
14.09
Average of return_3yr_pct
Average Alpha
1.25
Average of alpha
Average Beta
0.87
Average of beta
Top Fund Houses by 3y Return(%)
SBI Mutual Fund
HDFC Mutual Fund
ICICI Prudential MF
Nippon India MF
Axis Mutual Fund
Kotak Mahindra MF
fund_house
0
20
40
60
80
Sum of return_3yr_pct
Risk vs Return
return_3yr_pct
20
10
0.2
0.4
0.6
0.8
1.0
beta
category
--
ELSS
Flexi Cap
Gilt
Index
Index/ETF
Large & Mid Cap
Large Cap
Liquid
Mid Cap
Short Duration
Small Cap
Value
fund_house
--
Aditya Birla Sun Life MF
Axis Mutual Fund
DSP Mutual Fund
HDFC Mutual Fund
ICICI Prudential MF
Kotak Mahindra MF
Mirae Asset MF
Nippon India MF
SBI Mutual Fund
UTI Mutual Fund

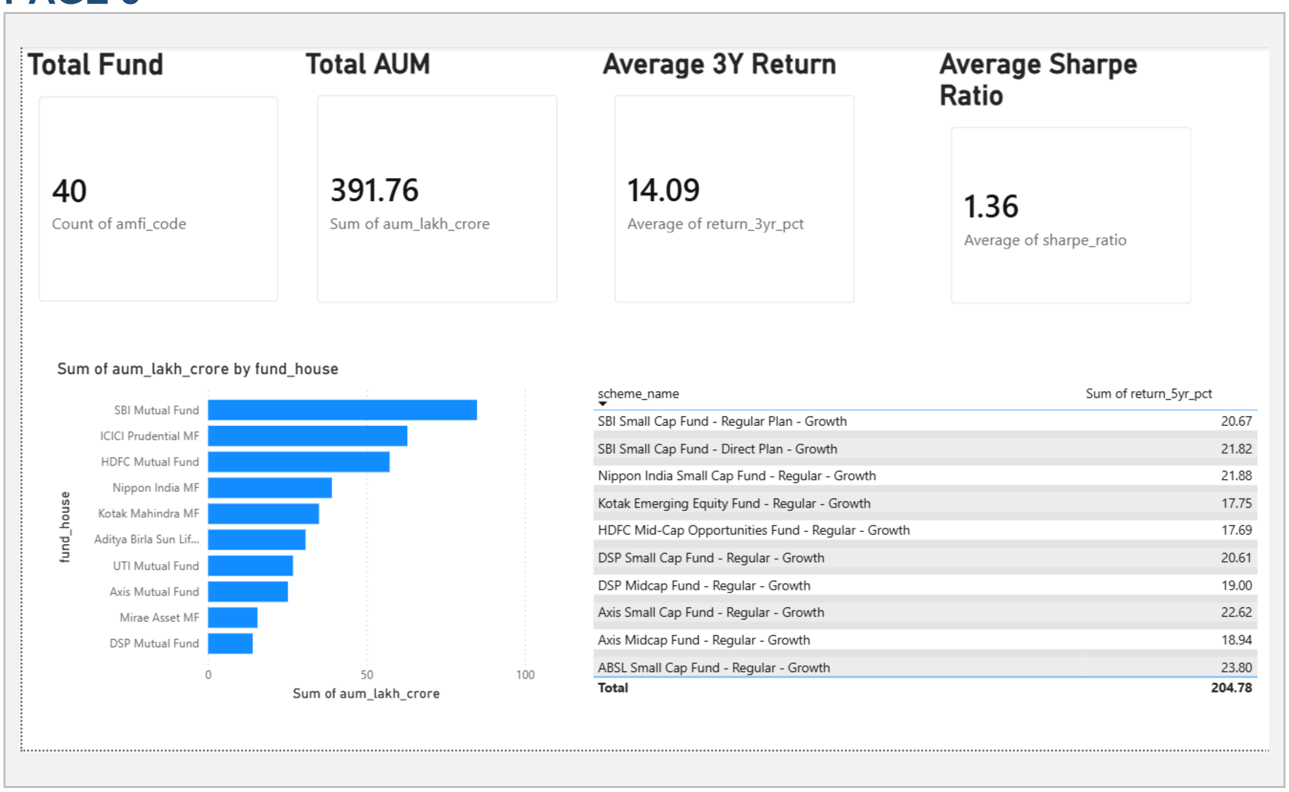
PAGE3


Investor & SIP Analysis Dashboard
Total Investors
5K
Count of investor_id
Total Investment Amount
4bn
Sum of amount_inr
Average SIP Amount
15.72
Average of sip_aum_lakh_crore
Total SIP Amount
91K
Sum of sip_inflow_crore
SIP Inflow Trend
Sum of sip_inflow_crore
31K
30K
29K
Oct 2025
Nov 2025
Dec 2025
Year
Lumpsum
Redemption
SIP
Transaction Amount by Type
transaction_type
Lumpsum
Redemption
SIP
0bn
2bn
Sum of amount_inr
category
--
Debt
Equity
Industry Folio Count
Sum of total_folios_crore
200
150
100
50
0
2022
2023
2024
2025
Year
city_tier and gender
B30
Female
Male
T30
Female
Male

PAGE4



PAGE 5



The Power BI dashboard provides an interactive view of mutual fund industry trends, investor behavior, performance metrics, and market insights, enabling effective business decision-making.

Key Findings

1. Mutual fund AUM has shown continuous growth over the analysis period.
2. SIP inflows indicate increasing retail participation in mutual funds.
3. Equity funds dominate investor preference compared to other categories.
4. Top-performing funds consistently outperform their benchmark returns.
5. Sector allocation shows a concentration of investments in a few major industries.
6. Correlation analysis identifies funds with similar performance behaviour.
7. Fund rankings help identify consistently high-performing schemes.
8. Benchmark indices have a notable influence on overall fund performance.

Conclusion

The Bluestock Mutual Fund Analytics Capstone Project successfully implemented an end-to-end analytics pipeline using Python, SQLite, SQL, and Power BI. Raw mutual fund datasets were transformed into meaningful business insights through a combination of data engineering, analysis, and visualization techniques.

The resulting dashboard provides a comprehensive view of industry trends, fund performance, investor behaviour, and market movements – enabling data-driven decision making for investors and stakeholders alike.

Future Enhancements

The following enhancements have been identified as potential next steps to extend the value and capability of this project:

- Real-time NAV updates
- Automated ETL scheduling
- Predictive fund return modeling
- Machine Learning integration
- Cloud deployment
- Interactive web dashboard